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CO–Ar continuum under the rotational band: can super-Lorentzian wings be involved?

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The CO–Ar continuum absorption was retrieved from spectrum recordings in the far-infrared range (40–160 cm^{-1}) using the Fourier spectroscopy technique with a synchrotron radiation source. The expected continuum pressure dependences were obtained. The continuum retrieved from the high-resolution spectra after subtracting the calculated resonant spectrum was consistent with the continuum measured at low resolution in the microwindows between lines. The observed continuum agrees well with the bimolecular absorption model constructed on the basis of the second virial coefficient and redistribution of integrated absorption from resonant CO lines to the continuum. We discuss the origin of the intensity depletion effect and propose a semiempirical model for the far wings which considers collision duration and is consistent with the experimental data.

1 Introduction

The development of self-consistent and physically based models of molecular absorption is in increasingly high demand for present-day atmospheric and climatic modeling.[1] Practical applications require unprecedented accuracy from the atmospheric radiation propagation models (see, *e.g.*, ref. 2 and references therein).

The absorption of an ideal gas can be accurately represented in terms of a sum of individual resonant lines, the spectral shape of which is characterized by a limited set of parameters conventionally catalogized in the spectroscopic databases for atmospheric and astrophysical applications. Actually, atmospheric gases can rarely be considered ideal. In most cases, intermolecular interactions affect the shape of resonant lines and give rise to the formation of underlying continuum absorption, the nature of which has been a challenging problem for many decades.[1,3] Nonideality of a real gas can be considered in terms of a virial expansion, *i.e.* by taking into account the contribution from molecular pairs, which very rarely needs to be supplemented by the allowance for triple or higher order molecular groups.[4] The spectroscopic signatures of molecular groups, such as true bound van der Waals or hydrogen-bonded dimers, are clearly established both in the so-called collision induced absorption and the water vapor continuum[5–8] under thermodynamic equilibrium conditions close to room temperature. The dimeric fingerprints in these spectra cannot be associated with absorption by individual monomers. One has to admit, however, that the first-principles rigorous discrimination among monomeric resonant and smooth absorption by molecular pairs is as yet unfeasible.

A real (non-ideal) gas absorption model requires first of all highly accurate knowledge of the line shape and other parameters for molecules of atmospheric and astrophysical interest. Also, knowledge of the non-resonant part of the absorption, the so-called continuum, is indispensable. Presently, the nature of the continuum is not yet established with satisfactory clarity. Researchers agree that it definitely results from intermolecular interactions. As the gas pressure increases, more and more monomers are getting involved into either stable or metastable dimers. Also, the role of molecular pairs which belong to the so-called free pairs cannot be neglected.[4] The appearance of true bound dimers may result in the formation of distinct peaks in the absorption spectrum. Moreover, the theoretical line shapes used in all propagation models are derived in the impact approximation, which is no longer valid at some distances from the line centers (the problem of the far line wings) leading to the uncertainty of continuum determination. As a result, semi-empirical continuum models are still widely used for applications (see, for example, ref. 9–12).

The water related continuum is, on the one hand, most demanded for applications because H_2O is the main greenhouse gas of the Earth's atmosphere. On the other hand, it is a challenge for theoretical description mainly due to the large permanent dipole moment of the H_2O molecule. Nevertheless, the mechanism of continuum absorption due to paired molecular states has much in common, if not unique, for all types of molecules. High-precision theoretical calculations of bimolecular

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absorption demonstrate good agreement with experimental data for molecular systems formed by non-polar molecules (N_2–N_2, N_2–Ar, CO_2–CO_2, and CO_2–Ar).[13–17] This suggests that the nature of the continuum absorption in these molecules is already well understood.

CO–Ar is one of the simplest molecular systems incorporating a polar molecule. This system is convenient for both experimental and theoretical studies of the continuum.[18] The resonant spectrum of the diatomic, weakly polar CO molecule is well known. Thus, the uncertainties of line shape parameters will have less impact than for other molecules. Accurate *ab initio* surfaces of potential energy and dipole moment are available for CO–Ar.[19–21] This system is much more amenable to theoretical calculations than systems including H_2O due to a higher rigidity of the CO molecule and a significantly smaller number of degrees of freedom. The CO–Ar continuum is much weaker than the H_2O related continuum but the difficulty of measuring it against the background of intense CO lines is facilitated by their regular structure.

This work seeks to retrieve the CO–Ar continuum in the range of the CO pure rotational band in support of subsequent theoretical calculations. However, measurements in the far-infrared range, where the band is located, are still difficult due to the lack of sufficient radiation sources and detectors.[22] The SOLEIL synchrotron is the unique option providing not only the high spectral brightness of radiation, but also the infrastructure and spectroscopic equipment. The obtained broad-band spectra allow the simultaneous determination of both the CO line parameters and the continuum, which makes the retrieval of the latter much more reliable (see the discussion section in ref. 23).

The objective of our study allows us to advance in one more direction. The experimental quantification of the resonant line integrated intensity depletion effect is presented in the recent paper.[12] The systematic decrease in the measured CO line intensities broadened by N_2 was observed and attributed to the super-Lorentzian behavior of the far line wings using classical molecular dynamics simulations (CMDS). In further work on the subject,[24] it was concluded that "This phenomenon is not modeled in the IUPAC-recommended Hartmann–Tran Profile,[25] which was derived based on the impact approximation which assumes instantaneous collision events". The intensity depletion effect was also demonstrated in pure CO_2.[23] At the same time, it was found that the CO_2 line intensity depletion strictly corresponds to the continuum intensity increase in agreement with the second virial coefficient (SVC) of the gas equation of state. This allowed an alternative interpretation of the continuum by CO_2 pair states. Two explanations of the same effect point to the inconsistency of the modern continuum nature understanding. This paper aims at the development of a method for the self-consistent description of the observed spectra in the CO–Ar system on the basis of a general approach[26] considering both (i) the formation of pair states in agreement with the SVC and (ii) the plausible behavior of the far line wings.

This article is structured as follows. Section 2 discusses briefly the experimental technique and conditions. The principles of data processing are reported in Section 3. The results of continuum retrieved are presented in Section 4, and their discussion and analysis are presented in Section 5.

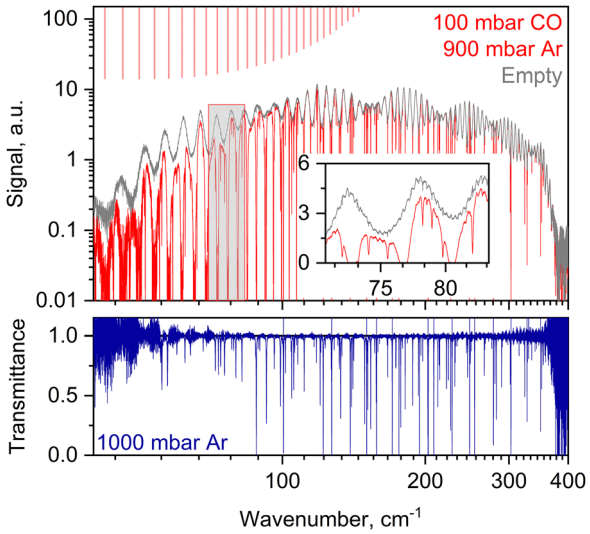

Fig. 1 A typical example of raw spectra recorded with an empty cell (gray curve) and with a mixture of 100 mbar CO and 900 mbar Ar (red curve). The vertical bars are the $^{12}C^{16}O$ spectrum stick-diagram. The lower panel shows the transmittance of 1000 mbar Ar.

2 Experimental

CO–Ar absorption data were recorded in the AILES beamline of the SOLEIL synchrotron. We used the standard synchrotron mode (500 mA) in the far infrared range providing the radiation power in about one order of magnitude higher than that of thermal radiation sources such as globar (see, for example, Fig. 1 in ref. 27). An additional bandpass filter was used in a liquid helium cooled detector. The spectra were recorded within 40–160 cm^{-1} using a Fourier transform spectrometer (Bruker IFS-125HR) equipped with a White type multi-pass cell (effective optical path length L = 151.75 ± 1.5 m). The gas temperature was 296.1 ± 0.4 K (permanently monitored by two PT100 sensors). Pure gases were supplied by Air Liquide. The intensity of the radiation transmitted through a cell with a CO–Ar mixture (the studied spectrum $I(\nu)$) or through an empty cell (reference spectrum or baseline $I_0(\nu)$) was recorded. The absorption coefficient was defined as $\alpha_{total} = (1/L)\ln(I_0(\nu)/I(\nu))$. (Hereafter, we call the absorption coefficient simply absorption for short.) Typical examples of $I(\nu)$ and $I_0(\nu)$ are shown in Fig. 1. In addition to CO lines, the enlarged excerpt shows intense lines of water vapor present as an unavoidable impurity. The partial water pressure was determined spectroscopically for each spectrum using an integral intensity of about 50 H_2O lines from HITRAN[28] and constituted about 0.2 ± 0.1 μbar.

A series of recordings were performed prior to the experiments to test the baseline stability. Pure Ar (non-absorbing in this spectral range) was used. The argon pressure was incrementally raised to 1000 mbar (the maximum possible pressure) and then reduced to 0. The lower panel of Fig. 1 shows that no

Table 1 Experimental conditions

Recording number	1	2	3	4	5
Number of pressure points	10	12	1	1	1
Filling CO pressure, mbar	100	150.1	99.82	149.7	50
Resolution, cm^{-1}	0.01		0.005		
Max total pressure, mbar	900.8		1000		
Number of acquisitions	100		1250		

baseline variations were observed with pressure variations. Diamond wedge-shaped windows of the gas cell (an average thickness of 0.9 mm) ensured the baseline stability but strongly affected the spectrum of transmitted radiation due to the high refractive index, leading to re-reflections of radiation and the appearance of standing waves with a period corresponding to the window thickness (Fig. 1). It is noted that traditional thin-film windows produce a smoother signal but cannot provide sufficient baseline stability.[27]

The experimental conditions are given in Table 1. Several series of spectrum recordings were obtained with different ratios of CO and Ar at low (0.01 cm^{-1}) and high (0.005 cm^{-1}) spectral resolutions. (The resolution of 0.005 cm^{-1} is comparable with CO line widths at the lower pressure of our recordings in the frequency range under consideration). Previous continuum studies in the AILES beamline (ref. 8 and 27 and references therein) show that the instrumental spectral function does not notably affect the broad continuum absorption retrieved between lines in microwindows of transparency. An excerpt of the resulting spectra is shown in Fig. 2.

Each series of the low-resolution spectrum recordings consisted of two independent pressure-sweeping stages. The first one started with pure CO at a certain pressure to obtain sufficient spectrum quality and continued by step-by-step adding of Ar up to a maximum pressure (gas filling stage). Then, the resulting mixture was gradually pumped out and a second series of spectra were recorded at each pressure step (gas pumping stage). The achievement of gas mixture equilibrium (several minutes) was controlled by analyzing the sequential fast scans recorded right after the pressure change. The gas pressure was measured using two Pfeiffer capacitance manometers (100 and 1000 mbar full scale) with a stated uncertainty of 0.25% of reading. The CO concentration and the mixing ratio in the sample were assumed to be constant at the first and second stages, respectively. In addition, three high-resolution spectra were recorded at several mixture ratios of CO and Ar.

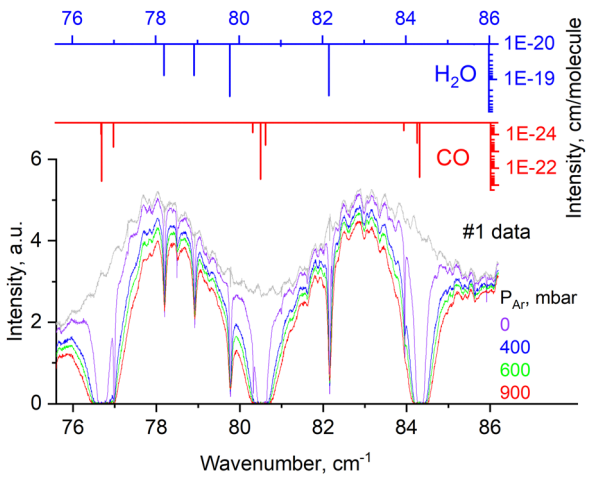

Fig. 2 Excerpts of the low-resolution spectrum series #1: empty cell (gray), 100 mbar of pure CO (purple), and with the addition of Ar up to 400, 600 and 800 mbar (blue, green and red, respectively). The stick-diagrams of CO and H_2O according to HITRAN data are shown above.

3 General approach to data treatment

The total absorption α_{total} was considered as the sum of the resonant lines of CO, α_{CO}, and H_2O, α_{H_2O}, and the continuum absorption of α_{cont} formed by the contributions of paired interactions between all gas mixture components (triple interactions under experimental conditions are negligible):

$$\alpha_{total}(\nu, P_{CO}, P_{H_2O}, P_{Ar}) = \alpha_{CO}(\nu, P_{CO}, P_{H_2O}, P_{Ar}) + \alpha_{H_2O}(\nu, P_{CO}, P_{H_2O}, P_{Ar}) + \alpha_{cont}(\nu, P_{CO}, P_{Ar}) \quad (1)$$

where ν is the wavenumber and P_{CO}, P_{H_2O} and P_{Ar} are the partial pressures of CO, H_2O and Ar.

The collisional broadening dominates for both CO and H_2O lines. The Doppler width of CO lines is almost an order of magnitude less than the minimum collisional width under experimental conditions. For H_2O, the Doppler line width is at least 5 times smaller than the collisional line width. The speed-dependence of line widths[29] was not observed as well. The resonance spectrum was modeled as a sum of lines with the Van Vleck–Huber profile,[30] supplemented by the first-order line mixing term to account for the effect of line interference, similarly to the Rosenkranz profile:[31]

$$\alpha_{Line} = n\frac{S}{\pi}\frac{\nu}{\nu_0}\frac{\tanh(h\nu/2k_BT)}{\tanh(h\nu_0/2k_BT)} \times \left(\frac{\Delta\nu_c + Y(\nu-\nu_0)}{(\nu-\nu_0)^2+\Delta\nu_c^2} + \frac{\Delta\nu_c + Y(\nu+\nu_0)}{(\nu+\nu_0)^2+\Delta\nu_c^2}\right) \quad (2)$$

where n is the concentration of absorbing molecules, k_B is the Boltzmann constant, h is the Planck constant, and T is the gas temperature. $\Delta\nu_c = \sum_i \gamma_i P_i$ is the collision width of the lines, γ_i is the broadening coefficient of the line by the i_{th} gas, and Y is the collision coupling (or mixing) parameter. The self-broadening coefficients, central frequencies ν_0, and integrated intensities S were taken from the HITRAN database.[28] The broadening coefficients of CO lines by argon pressure with a rotational quantum number of $J < 15$ were taken from ref. 32, which are consistent with the supporting *ab initio* calculations for $J < 22$ from the same work and with the high accuracy measurement results[18] for the R0 and R1 lines within their error bars (the difference is 0.4 and 0.2%, respectively). For higher J, we used the broadening coefficients obtained in this work from the high-resolution spectra (see Appendix A for details).

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The water vapor pressure was at least 2 orders of magnitude less than the CO and Ar pressures; thus, broadening of CO lines by H_2O pressure was neglected. The effective widths of H_2O lines in the high-resolution spectra were determined based on the fit of the lines similar to CO lines. For low resolution spectra, the HITRAN air broadening coefficients multiplied by 1.35(27) were used for the H_2O–CO broadening. A uniform value of 0.026(10) cm^{-1} per atm was applied for all H_2O lines for their broadening by Ar, which is consistent in magnitude with data from ref. 33 in the frequency range of 1864–2115 cm^{-1}. The collision coupling effect was neglected for water lines.[34] For CO lines, the mixing parameters were calculated as $Y = y_{CO}P_{CO} + y_{Ar}P_{Ar}$ with y_{CO} and y_{Ar} from ref. 18.

The line profile (2) was derived in the impact approximation and valid within $\pm 1/(2\pi\tau_c)$ around the line center, where τ_c is the collision duration.[35] For CO–Ar collisions, τ_c is about 0.9–1.7 ps depending on the CO rotational state[1] and, therefore, the line shape is valid for frequency detuning not exceeding several cm^{-1}. However, at the first stage of spectrum modeling, the line wings were cut off at the frequency detuning from the line center ν_{cut} = 25 cm^{-1}, which is considered to be the standard.[36,37] The influence of the ν_{cut} value on the retrieved continuum is discussed in Appendix B along with the other mechanisms contributing to its uncertainty. It should be noted that the resonance line profile was not lowered down after the wing cutting (as was done in the MT_CKD approach[38]), and the "pedestal" under the lines was considered as a part of the resonance spectrum to preserve its integrated intensity.

The continuum was retrieved from eqn (1) as the difference between the total observed absorption α_{total} and the modeled resonant absorption α_{CO} and α_{H_2O}.

For the continuum interpretation, let us consider the equation of gas state in a virial form:

$$\frac{PV_m}{RT} = 1 + \frac{B(T)}{V_m} + \frac{C(T)}{V_m^2} + \ldots, \quad (3)$$

where R is the universal gas constant, V_m is the molar volume, and $B(T)$ and $C(T)$ are the second and third virial coefficients characterizing the "efficiency" of double and triple interactions. It can be shown with data from ref. 39 that $B(T)/V_m$ is about 2 orders of magnitude greater than $C(T)/V_m^2$ for a mixture of CO and Ar at room temperature. This shows that the consideration only of pair interactions is sufficient in our case. The conclusion is also confirmed by the fact that the mean free path of CO molecules between collisions is 100–500 times larger than the radius of the effective collisional cross section under our experimental conditions, according to the estimates based on gas kinetic theory equations and widths of the CO lines from the HITRAN2020 database.

Assuming that the H_2O–CO continuum is comparable to the water self-continuum, and the H_2O–Ar continuum is comparable to the H_2O–N_2 continuum (which is an obvious overestimation in both cases), one can expect the total H_2O-related continuum in our case to be 3 orders of magnitude smaller than the experimental noise. Thus, all observed continuum absorption can be attributed to the CO self-continuum α_{CO-CO} and the CO–Ar foreign continuum α_{CO-Ar} as follows:

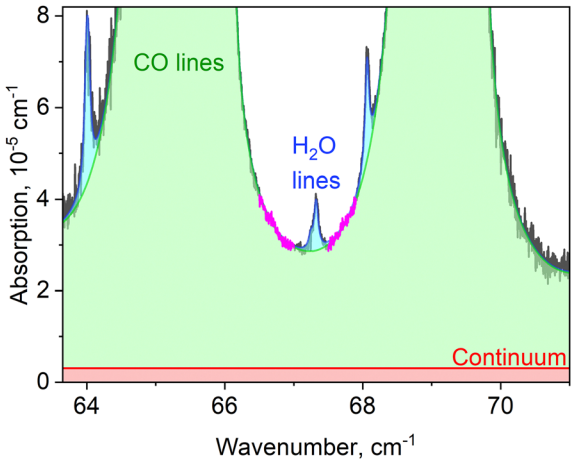

Fig. 3 Total absorption spectrum fragment (black curve) for 150 mbar CO and 750 mbar Ar. The CO resonant spectrum contribution is shown in green, H_2O impurity lines are in blue, and the continuum absorption is in red. Microwindows for determining the continuum are highlighted in pink.

$$\alpha_{cont}(\nu, P_{CO}, P_{Ar}) = \alpha_{CO-CO}(\nu, P_{CO}, P_{Ar}) + \alpha_{CO-Ar}(\nu, P_{CO}, P_{Ar}) = C_{CO-CO}(\nu)P_{CO}^2 + C_{CO-Ar}(\nu)P_{CO}P_{Ar} \quad (4)$$

where C_{CO-CO} and C_{CO-Ar} are continuum absorption cross sections (cm^{-1} $mbar^{-2}$).

4 Continuum cross-section retrieval

C_{CO-CO} and C_{CO-Ar} were determined separately for each series of experimental recordings. The continuum retrieval is more reliable when the parameters of the resonance lines are also fitted.[18,23,40] The high-resolution spectra allow such processing while the low-resolution spectra are suitable for determining the continuum cross sections using *a priori* knowledge of the continuum pressure dependence (4). This was done in as many as possible microwindows of transparency between CO and H_2O lines, where the influence of the line parameter uncertainty is minimal, and the instrumental resolution does not affect the observed absorption. Fig. 3 illustrates the choice of the microwindows and the continuum contribution to the observed absorption. The continuum absorption $\alpha_{cont}(\nu, P_{CO}, P_{Ar})$ was calculated in the frequency range of each chosen microwindow using eqn (1) and the resulting value was averaged over frequency.

A. Low resolution spectra

The data for increasing and decreasing pressures were treated separately (series #1 and #2).

During the gas filling stage, P_{CO} is assumed to be constant and α_{cont} depends linearly on Ar pressure according to eqn (4). Such behavior is confirmed in Fig. 4. It should be noted that the deviation of the points from the expected linear dependence is about an order of magnitude smaller than their uncertainty due

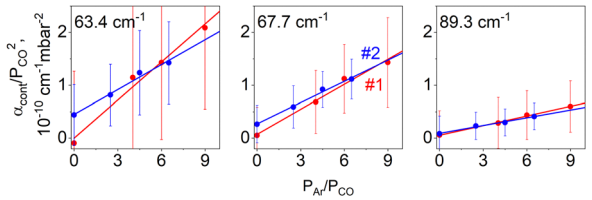

Fig. 4 Derived α_{cont}/P_{CO}^2 *versus* P_{Ar}/P_{CO} for series #1 (red) and #2 (blue) for increasing pressures in three typical microwindows. The dots are experimental data, and the lines correspond to their approximations by eqn (5).

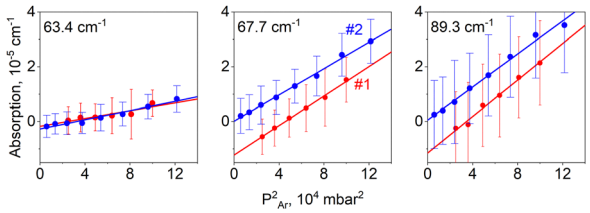

Fig. 6 Continuum absorption α_{cont} *versus* Ar pressure squared obtained from decreasing pressure data for series #1 (red) and #2 (blue) in three typical microwindows. The dots correspond to the experimental data, and the lines are their approximation by eqn (6).

to the experimental noise level (Fig. 3). C_{CO-CO} and C_{CO-Ar} were determined for each microwindow as a linear function coefficient fitted to the experimental data:

$$\frac{\alpha_{cont}(\nu, P_{Ar})}{P_{CO}^2} = C_{CO-CO}(\nu) + C_{CO-Ar}(\nu)\frac{P_{Ar}}{P_{CO}}. \quad (5)$$

C_{CO-CO} can also include a potential baseline shift due to temperature drifts or due to variations in the power of the radiation passing through the cell. These parasitic effects would prevent the C_{CO-CO} reproducibility when the ratio of CO and Ar pressures was changed and a series of measurements were carried out in different time periods. The agreement between C_{CO-CO} obtained from series #1 and #2 within the uncertainty interval confirms that the impact of baseline variations is almost negligible. It is also supported by the agreement of C_{CO-CO} retrieved by the linear fit of pressure dependence and from pure CO spectrum recordings (the first pressure points in series #1 and #2), although the uncertainty of the self-continuum determined from a single low-resolution recording is about 2 times greater than the uncertainty from the full dataset (Fig. 5). The uncertainty of the obtained C_{CO-CO} takes into account the noise level in the absorption spectra, the fit uncertainty and the data deviation from series #1 and #2 (the errors associated with the uncertainty of temperature and pressure readings are more than an order of magnitude smaller and may be neglected). The CO–CO component is 5–15% of the observed continuum, depending on the ratio of CO and Ar pressures.

During the gas pump-out stage, the P_{CO}/P_{Ar} ratio was considered to be constant. Eqn (4) can be modified as follows:

$$\alpha_{cont}(\nu, P_{CO}, P_{Ar}) = \Delta\alpha(\nu) + \beta(\nu)P_{Ar}^2$$

where

$$\beta(\nu) = C_{CO-CO}(\nu)\left(\frac{P_{CO}}{P_{Ar}}\right)^2 + C_{CO-Ar}(\nu)\frac{P_{CO}}{P_{Ar}}, \quad (6)$$

(which is similar to eqn (7) in ref. 27).

The expected quadratic dependence of the continuum on Ar pressure was observed in all microwindows. Three typical examples are shown in Fig. 6. The extrapolation of absorption data to zero pressure (Fig. 6) reveals the baseline shift accounted for by $\Delta\alpha(\nu)$. The CO–Ar continuum cross section $C_{CO-Ar}(\nu)$ was determined in each microwindow from the slope of these dependences $\beta(\nu)$ with $C_{CO-CO}(\nu) = 1.46 \times 10^{-10}\nu^{2.483}e^{-0.05\nu}$ obtained as an empirical approximation of the CO–CO cross sections retrieved from the increasing pressure stage data (green line in Fig. 5).

$C_{CO-Ar}(\nu)$ obtained from the data at increasing and decreasing pressures agree within 3σ uncertainty which takes into account the fit of $k(\nu, P_{CO})$ and $\beta(\nu)$, with the weight inverted to the noise level in each microwindow (left and middle panels in Fig. 7). This allows them to be averaged to reduce the impact of potential systematic errors. The right panel in Fig. 7 shows

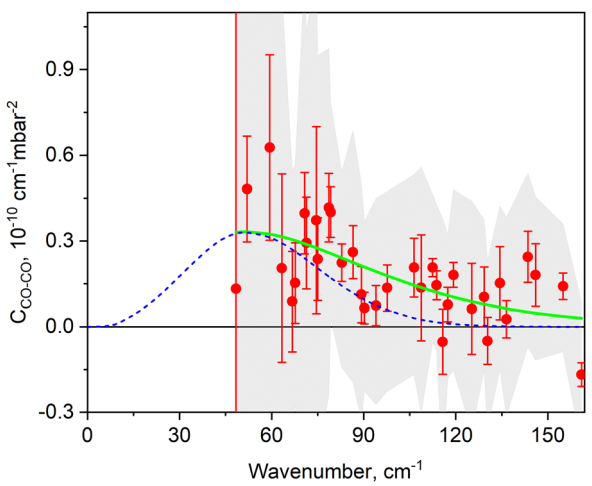

Fig. 5 CO–CO continuum cross section determined from series #1 and #2 (red). The green curve is approximation of experimental points. The gray area corresponds to the range of values C_{CO-CO} determined from two pure CO spectra, including their uncertainties. The blue dotted curve is the CO–CO bimolecular absorption model (Section 5).

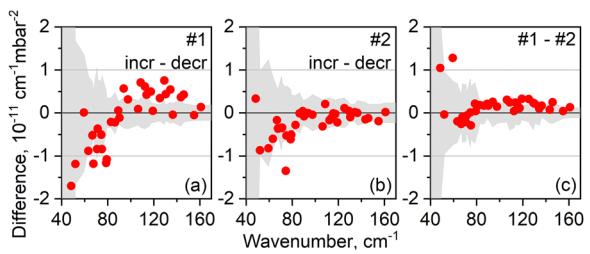

Fig. 7 The difference between $C_{CO-Ar}(\nu)$ obtained from increasing and decreasing pressure data in series #1 (panel a) and #2 (panel b) and between the averaged $C_{CO-Ar}(\nu)$ for series #1 and #2 (panel c) is shown by red dots. The gray area indicates the 1σ uncertainty.

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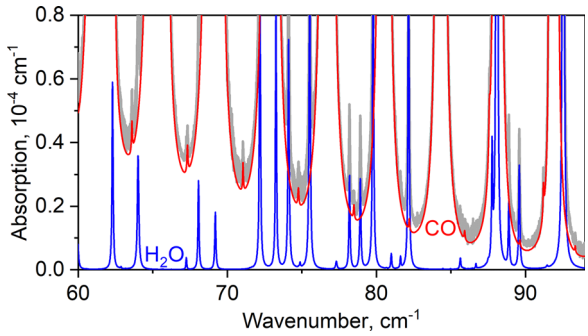

Fig. 8 Excerpt of spectrum #3 (gray) and the calculated contributions of CO (red) and H_2O (blue) resonant lines.

good agreement of averaged $C_{CO-Ar}(\nu)$ obtained from series #1 and #2, which also allows their averaging for further consideration. The uncertainties of the retrieved CO–Ar continuum absorption are associated with the experimental noise, the uncertainty of the fit of $k(\nu, P_{CO})$ and $\beta(\nu)$ coefficients, and the deviations between different datasets (increasing and decreasing pressures, series #1 and #2). They were considered to be independent and thus the total uncertainty was calculated as the root of the sum of their squares. The other sources of possible errors (the uncertainties of CO and H_2O line shape chosen for resonant spectrum modeling including uncertainties of ν_{cut}, S, γ_i, and Y) are discussed in Appendix B.

B. High resolution spectra

High-resolution spectra #3–#5 (Table 1) were used for direct continuum determination from eqn (1). The resonance line parameters were obtained from the fit of the same spectra (details are given in Appendix A). An excerpt of spectrum #3 and models of CO and H_2O resonant spectra are shown in Fig. 8.

The self-continuum component $C_{CO-CO}(\nu)$ was considered similar to the low-resolution spectra processing. Fig. 8, like Fig. 3, shows that the continuum, that is the systematic difference between calculated and measured absorption, is extremely small. It is hardly noticeable even between lines. However, $C_{CO-Ar}(\nu)$ obtained from spectra #3–#5 agrees within the noise level. This agreement indicates the reliability of the result and allows averaging the data. Fig. 9 shows that the result is in good agreement with the data obtained from low-resolution spectra.

The smaller spread of points above 120 cm^{-1} around the zero level in the absence of strong CO lines is noted. Therefore, although the influence of baseline instability cannot be completely excluded, we can suggest that it is less than other uncertainties.

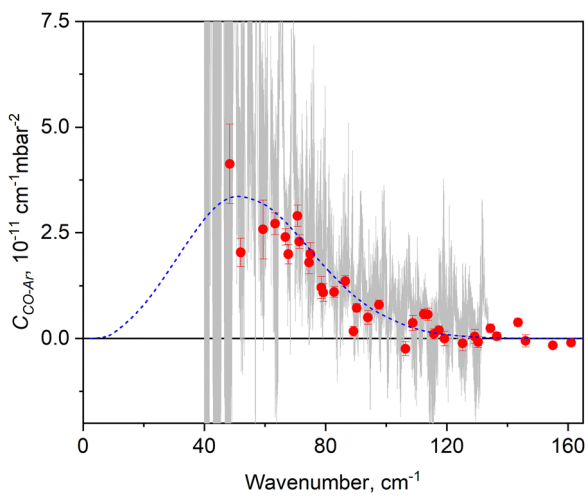

Fig. 9 CO–Ar continuum cross section $C_{CO-Ar}(\nu)$ obtained from high-resolution (gray) and low-resolution (red) spectra. The blue dotted curve is the CO–Ar bimolecular absorption model (Section 5).

5 Discussion

The indissoluble connection between the continuum and the resonance spectrum has been repeatedly discussed in the literature. The latest confirmation of the statement was experimentally demonstrated by Reed *et al.*[12] and Birk *et al.*[23] We used the general results of these works for interpreting the observed continuum. Thanks to the high sensitivity of the CRDS technique, the authors of ref. 12 experimentally observed a reduction of the CO line integrated intensity with increasing pressure. The reduction is rather small but systematic and may be significant taking into account high modern requirements to the spectroscopic data accuracy and thus requires appropriate corrections in propagation models. The manifestation of the effect, called "intensity depletion", was confirmed by CRDS. It looks like an intensity transfer from the maximum of the resonance lines to the region of their wings. In ref. 12, this effect was associated with a rapid change in the autocorrelation function of the CO dipole moment at the initial moment of time ("incomplete collisions"), leading to the super-Lorentzian behavior of the far wings of the lines.

It is noted that the intensity depletion effect was observed long before (see, for example, ref. 41,42 and references therein) and qualitatively interpreted (including its J-dependence) in terms of bimolecular absorption.[43–45] Following the latter works, we performed a numerical simulation of the observed effect using very basic information about the compressibility of gases. Recall that the higher the gas pressure, the more the difference of the gas from its ideal model affects the measurement result. The real gas pressure increases slightly slower than its concentration due to the sticking of the monomer molecules to each other and forming stable (bound) and metastable (quasi-bound) double molecules or dimers. (Note that by the monomer we assume the molecule moving from one collision to another and interacting with the field as a single object.) The authors of ref. 12 considered the gas non-ideality by introducing the compressibility factor $Z(T,P)$ into the CO number density ($n_a = x_{CO} \times n_i(T,P)/Z(T,P)$, where x_{CO} is the CO mole fraction and n_i is the ideal-gas number density) which increases the concentration of real gas molecules compared to an ideal gas. The molecules trapped in the form of bound or quasi-bound dimers form new absorbing objects in a gas and cannot contribute to the integrated intensity of the monomer resonant lines, since the dimer dipole moment and energy levels are

determined by the compound of both monomers, so that the spectra of dimers and monomers are entirely at variance. Only the single monomer molecules should be taken into account for experimental determination of line intensities.

The number of dimers and the corresponding change in the number of monomers in gas can be evaluated from the SVC, following the ideas of ref. 4, keeping in mind that the number of dimers in the gas is significantly less than the number of monomers (*i.e.*, the pressure is not very high, and the gas is almost ideal, which is valid for the experimental conditions used in ref. 12 and in our work). Within these assumptions, eqn (3) reduces to:

$$P = nkT + B(T)\frac{n^2kT}{N_a}, \tag{7}$$

where the first term corresponds to the pressure of an ideal gas, and the second one is the correction associated with the formation of bimolecular states. Both monomers and the formed dimers contribute to the total gas pressure ($P = P_m + P_d$, $P_m = n_m kT$, and $P_d = n_d kT = K_2P^2$, where the indexes m and d are for monomers and dimers, respectively, and K_2 is the equilibrium constant or dimerization constant associated with both the bound and quasi-bound states). The total concentration of molecules in the gas is $n = n_m + 2n_d$ (two monomers are required to form a dimer), and its difference from an ideal gas Δn is:

$$\Delta n = 2K_2n^2kT. \tag{8}$$

In the case of a mixture of two gases, two more terms will appear in eqn (8) for dimers from the second gas and for heterodimers (*i.e.* dimers consisting of different monomers) of the form $2K_{2_{12}}n_1n_2kT$, where $K_{2_{12}}$ is the equilibrium constant of heterodimers, and n_1 and n_2 are the concentrations of gases in the mixture. It follows from this equation that the relative change in the concentration of absorbing molecules $\Delta n/n$ in a pure gas ($n = n_1 = n_2$) and in a gas mixture with a constant relative composition (n_1/n_2 = const) linearly depends on the gas pressure (with the same degree of accuracy as the linearity of the collision parameters of the line shape). This linearity was confirmed experimentally (at $n_{CO} \ll n_{N_2}$) and theoretically ($n_{CO} = n_{N_2}$) in ref. 12.

For the quantitative assessment of the effect, we need equilibrium constants. Here, they are obtained from the expression $B(T) = b_0 - K_2 \cdot kT^4$ (where b_0 is the excluded volume estimated from the known critical parameters of the gas) using the $B(T)$ data from ref. 39. For the CO and N_2 mixture, the relative change in the concentration of single CO monomers and, consequently, the integrated intensity of its resonant lines at room temperature is about 4% MPa^{-1}. This is in good agreement with the relative intensity correction for a R1 line of 2.9(3)% MPa^{-1}.[12] It is noted that our estimate does not include the dependence on the rotational state of the molecule and is just an effective value for the gas. However, the lower the value of the rotational quantum number J, the easier the molecules stick to each other during the collision,[43,45] which explains the dependence of intensity depletion on J observed in ref. 12.

The intensity depletion effect in CO_2 spectra was experimentally confirmed in ref. 23. Both the linearity of the relative intensity decrease with pressure and the dependence of the magnitude of the effect on the rotational state were demonstrated. We estimated the change in the CO_2 lines integrated intensity using eqn (8) and the $B(T)$ data from ref. 39. The obtained values (0.014 and 0.036 atm^{-1} for temperatures of 296 and 212 K, respectively) are in good agreement with the measurement results (Fig. 11 and 12 in ref. 23).

In addition, the authors of ref. 23 showed that the integrated intensity of the continuum arising with the pressure increase corresponds exactly to the summed decrease of the integrated intensity of monomer resonant lines. This coincidence confirms the assumption made in ref. 43 and indicates the following. (I) A CO_2 molecule interacts with the field with the same efficiency both for a single monomer molecule and for a molecule in the pair state. Thus, the power of the absorbed radiation is the same in both cases. (II) The magnitude of the induced dipole moment during the collision is significantly less than the permanent one, and the absorption corresponding to the induced dipole can be neglected. This is consistent with the fact that the absorption associated with free molecular pairs of water vapor produced by an induced dipole moment is negligible in comparison with the dimer absorption largely determined by a permanent dipole.[46–49] We simulate the CO–Ar continuum spectrum assuming that these conclusions are valid for this collisional system as well.

Recall that the spectra of pair states contain a large number of unresolved (uniformly broadened) lines merging into wide bands of the so-called collision-induced absorption (CIA). This is due to the short lifetime of metastable states and the transient dipole moment of free molecular pairs, as well as to the significantly higher number of degrees of freedom of a double molecule compared to its parent monomers. It is well known that the shape of the CIA bands roughly resembles the envelope of the resonance spectrum of a monomer, which is observed as dipole transition lines for polar monomers or quadrupole transition lines for nonpolar ones (see, for example, Fig. 3 from ref. 7, or Fig. 6 from ref. 50).

We approximated the envelope of the CO pure rotational spectrum in the 0–200 cm^{-1} range (modeled using HITRAN data) by a smooth empirical function. The function normalized by its area has the form $6.434 \times 10^{-7}\nu^3 \exp(-8.64 \times 10^{-4}\nu^{1.9})$ and is expressed in units of cm, provided that the frequency ν is in cm^{-1}. The integrated intensity of our continuum model was found as the product of the total intensity of the rotational band lines (1.842×10^{20} cm mol^{-1}) and the change of the concentration of single monomers due to dimerization Δn calculated from eqn (8) and (9) using the SVC for CO–CO and CO–Ar from ref. 39 and the corresponding critical parameters of gases. The result is shown in Fig. 5 and 9 for CO–CO and CO–Ar, respectively. The almost perfect agreement of the proposed model with the experimental data confirms that the continuum observed under the CO rotational band is due to the formation of pair molecular states similar to the continuum under the CO_2 bands observed in ref. 23.

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Our analysis would be incomplete without addressing the issue of the far wings of resonant lines. First of all, we would like to point out the different meaning of the terms "Super-Lorentzian" and "Sub-Lorentzian" wing in the literature. Initially, they were used to interpret the excessive absorption between resonant lines within rovibrational bands and, respectively, insufficient absorption between the bands compared to the sum of Lorentzian profiles of the lines from these bands. The authors of ref. 37 assume that the total absorption, including the continuum, can be represented as a sum of resonant line profiles with modified wings. They use sub- and super-Lorentzian wings to interpret the result of empirical modeling of the water vapor spectrum. In ref. 12, the term "super-Lorentzian behavior" has the same meaning, leaving the nature of the excess absorption outside the scope of the study.

An alternative meaning of these two terms was proposed in ref. 26 (more details can be found in ref. 51), where it was referred to the shape of resonant lines of monomers absorbing radiation separately rather than forming pair states. The interaction between monomers was considered as an action of an external force, smoothly stopping the free oscillations of the monomer dipole moment.

The smoothness of the external force action during collision on the molecule–oscillator and the consequence of this effect was theoretically considered in ref. 36. The requirement of the temporal continuity of the autocorrelation function of the dipole and its derivatives leads to an exponential (*i.e.* Sub-Lorentzian) decay of the monomer line wing at large detuning from the line center. For the super-Lorentzian behavior, the molecule must have sufficient time to get additional energy from the collision partner before losing the initial coherence with the field. This can be accomplished, for example, due to the mutual rotation of molecules during the collision (see Fig. 4 in ref. 26). The effective collision duration τ_c and a number of CO turns during the collision with Ar at room temperature can be estimated from gas kinetic equations. (An effective collision cross section is retrieved from the pressure broadening coefficient of the CO line having the maximum intensity within the rotational band.) Such consideration shows that the CO molecule makes less than three turns during the collision. Taking into account the necessity of averaging this process over the gas volume, the super-Lorentzian behavior is unlikely for CO lines. This is consistent with the fact that our continuum model based on the estimate of the number of dimers agrees with the observations and does not require additional absorption mechanisms. For comparison, an H_2O molecule can make up to 35 turns colliding with each other according to a similar estimation and the water continuum cannot be explained by dimers alone.[7,8,26,40]

To simulate a CO monomer's resonant line shape including the far wings, we use eqn (17) from ref. 26 for the shape of a train of molecular dipole oscillations with $A = 0$ (thus excluding the super-Lorentzian behavior):

$$F(t) = \frac{1}{2}\left(\tanh\left(\frac{4t + 2\tau}{\tau_c}\right) - \tanh\left(\frac{4t - 2\tau}{\tau_c}\right)\right), \qquad (9)$$

where τ is the free path time. The corresponding line shape is obtained after the Fourier transform and gas averaging of such oscillations (eqn (14) and (16) in ref. 26). The result for $\tau_c = 1$ ps (recall that for CO–Ar the collision duration is 0.9–1.7 ps[1]) is shown in Fig. 10 together with its Lorentzian counterpart $L(\nu)$. The obtained profile was approximated by the empirical function

$$\alpha^{F}_{Line}(\nu) = L(\nu)e^{-\frac{(\nu-\nu_0)^2}{\Delta\nu^2}}, \qquad (10)$$

where $\Delta\nu$ is the width of the Gaussian function, which was found to be equal to 25 cm^{-1}.

The line shape function (10), with $L(\nu)$ replaced by $a_{Line}(\nu)$ from eqn (2), was applied to retrieve the CO–Ar continuum from the series of spectra #2 in the same way as the truncated profile in Sections 3 and 4. The result is shown in Fig. 11. In all microwindows, its difference from the original one is less than the statistical error. However, the physically justified consideration of τ_c makes the continuum more "purified" from the monomer absorption than using the truncated profile with, in fact, arbitrarily chosen ν_{cut}. The good quantitative agreement

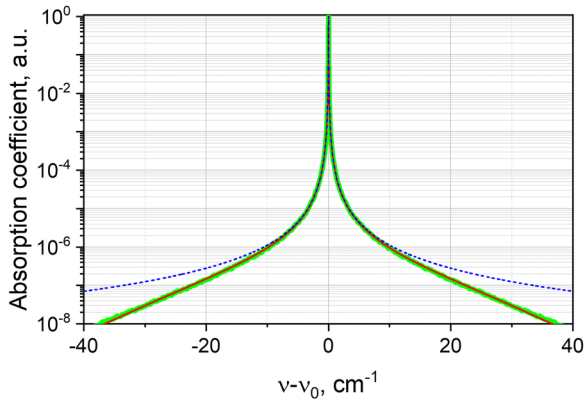

Fig. 10 The line profile numerically calculated from eqn (9) (green) and its approximation by function (10) (red). The blue dotted curve is the $L(\nu)$ profile.

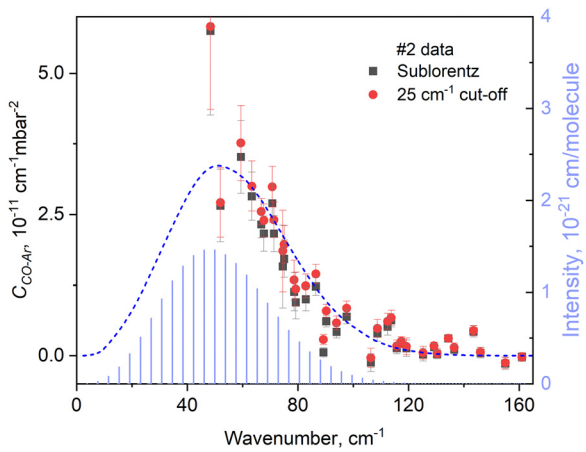

Fig. 11 $C_{Co-Ar}(\nu)$ obtained from synchrotron spectra of #2 series using a line shape model with the wings cut off at 25 cm^{-1} from line centers (gray squares) and modified using function (10) (red circles). The blue line shows a model of CO–Ar bimolecular absorption shaped by the envelope of its resonance spectrum.

of the continuum obtained with our model indicates the bimolecular nature of this absorption.

Note that our semi-empirical models for the bimolecular absorption and the extended shape of the single monomer line agree with the observations and allow us to consistently separate the absorption mechanisms in gases. However, they are still waiting for theoretical confirmation from the first principles. The classical trajectory simulation (CTS) method seems to be most promising since it enables detailed monitoring of the evolution of the dipole moment of a monomer during collision.[52] The efficiency of this method for calculating the bimolecular spectra of nonpolar molecules was demonstrated in ref. 15–17. The detailed agreement between the calculated and observed spectra obtained in these works allows us to state that the continuum nature has been established, at least for the studied collisional systems.

Conclusions

The CO–Ar continuum absorption has been revealed in the range of the CO rotational band using the Fourier transform spectroscopy technique with a synchrotron radiation source. The reliability and self-consistency of the data have been confirmed by the expected pressure dependence and by the agreement between different methods of continuum retrieval from high- and low-resolution spectra.

Based on the experimental results from ref. 23, we have proposed a hypothesis according to which the efficiency of interaction with the field is identical for a single monomer molecule with a permanent dipole moment and for the same molecule in a pair state. This enables construction of a quantitative semi-empirical model of bimolecular absorption in CO–CO and CO–Ar collisional systems using data about the SVC and the critical parameters of these gases. The model allowed us to reach the agreement with the experimentally retrieved continuum using both the traditional modeling of the resonant profile with cut-off far wings and the extended semi-empirical line shape with physically justified collision duration.

We also used the developed model to interpret the experimental data from ref. 12 and 23. Our estimates of the relative change in the monomer concentration due to the dimer formation in pure CO_2 and in a mixture of CO with N_2 using SVC and critical gas parameters are well consistent with the above observations. This confirms the theoretical interpretation[43,45] of the intensity depletion effect as a bimolecular absorption in gases. The effect of line intensity depletion is a natural part of molecular absorption in gases and as such it has to be properly taken into account in upcoming propagation models.

In conclusion, let us come back to the discussion whether the IUPAC-recommended Hartmann–Tran profile[25] requires some modification taking into account the intensity depletion effect (as pointed by the authors of ref. 24). Our work has demonstrated that the effect is not related to the line profile of single monomers; therefore, the Hartmann–Tran profile can be used without modifications for resonant absorption modeling considering an appropriate number density of the single monomers (which obviously decreases when some monomers form dimers) contributing to the absorption.

Author contributions

A. O. Koroleva: visualization, validation, methodology, software, formal analysis, and writing – original draft. T. A. Galanina: writing – review and editing, visualization, formal analysis, and data curation. M. A. Koshelev: writing – review and editing, data curation, and validation. M. Yu. Tretyakov: writing – review and editing, validation, data curation, supervision, project administration, methodology, and conceptualization.

Data availability

The data that support the findings of this study are available within the article.

Conflicts of interest

There are no conflicts to declare.

Appendices

Appendix A

CO–Ar collisional broadening coefficients γ_{CO-Ar} were retrieved based on the fit results of 24 $^{12}C^{16}O$ lines from 3 high-resolution spectra (#3–#5). The apparatus function width (about 2×10^{-3} cm^{-1} for 0.005 cm^{-1} resolution) was more than 13 times less than the fitted line widths and thus the impact of the apparatus function was neglected. The self-broadening component was subtracted using HITRAN2020 data. The accuracy of the obtained γ_{CO-Ar} was limited due to the large optical depth

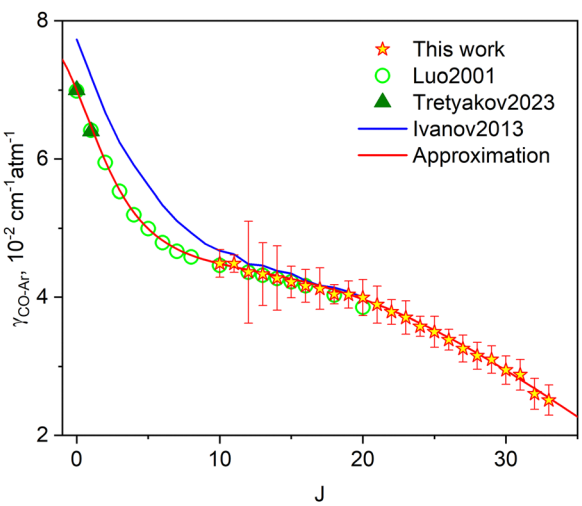

Fig. 12 Pressure broadening coefficients γ_{CO-Ar} obtained in this work (asterisk) with their fit uncertainty in comparison with previous measurements[18,32] (green open circles and triangles) and CTS calculations (blue curve). The Padé approximation for data used in this paper is shown by the red curve.

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saturation of intense lines near the maximum of the CO rotational band. The agreement within 5% between the coefficients obtained from different spectra and the agreement with the previous measurements[32] for $10 < J < 19$ indicates the reliability of our data (Fig. 12). Compliance within 15% with the results of the CTS calculations from ref. 53 should also be mentioned.

The experimental results from ref. 32 were applied for $J < 19$. For rotational lines with higher J, the coefficients obtained in this paper were used. For the continuum retrieval, the following Padé approximation of these data was employed:

$$\gamma_{\mathrm{CO-Ar}}(J) = \frac{0.0697(3)\cdot\left(1+0.0245(74)J^2-1.7(8)\times10^{-7}J^5\right)}{1+0.072(9)J+0.036(12)J^2+3.8(8)\times10^{-7}J^5}\ \mathrm{cm}^{-1}\ \mathrm{atm}^{-1}. \quad (A1)$$

Appendix B

The uncertainties of the retrieved CO–Ar continuum absorption are associated with the experimental noise and with the data processing method (including, in particular, the deviation of the observed absorption pressure dependence from the expected one). The other sources of possible errors are associated with the uncertainties of CO and H_2O line shapes chosen for resonant spectrum modeling including the uncertainties of ν_{cut}, S, γ_i, and Y. (Influence of the higher order line mixing parameters to line intensities[54] is expected to be within intensity uncertainties). To estimate their impact on the observed continuum, we varied the ν_{cut} from the values corresponding to the applicability of the impact approximation up to the unrealistically high values corresponding to nearly infinite Lorentzian line wings. The other line shape parameters were changed (simultaneously for all lines) from minimum to maximum within their uncertainties taken from original studies or from HITRAN error codes. The subsequent change in the continuum magnitude absorption reflects its uncertainty associated with the varied parameter of the resonance spectrum.

None of the performed modifications of the CO spectrum led to the continuum absorption shape change. A decrease/increase in the line intensities, broadening and mixing coefficients results in an increase/decrease in the continuum magnitude. Quantitative estimates of these changes are given in Table 2. The sign of the continuum change indicates whether it increases or decreases. Thus, when ν_{cut} increases to 50 cm^{-1}, the continuum becomes zero, and negative at higher ν_{cut}. The influence of self-broadening and Ar-broadening coefficient uncertainties ($\gamma_{\mathrm{CO-CO}}$ and $\gamma_{\mathrm{CO-Ar}}$) was evaluated jointly, since they equally affect the line shape. It is noted that the 1σ-uncertainty of the continuum is about 20% of its value. It is also noted that the estimates of HITRAN uncertainties are rather conservative (as follows from the water foreign-continuum study[27,55]).

Therefore, the uncertainties of the continuum associated with S_{CO} and $\gamma_{\mathrm{CO-CO}}$ are probably closer to their upper limit.

Despite the small H_2O concentration and the choice of microwindows far from H_2O lines, the water resonant spectrum contribution to the observed absorption is comparable to the continuum uncertainties. In the case of low-resolution spectra, it is partially reduced by the $\Delta\alpha(\nu)$ correction in formula (6) (we can subtract the permanent absorption component associated with H_2O). However, the H_2O partial pressure, determined from known line integrated intensities, changes almost linearly with increasing Ar pressure either due to the presence of some H_2O in our Ar bottle or due to adsorption/desorption processes. Neglecting the contribution of H_2O, the absorption spectrum in selected microwindows increases the value of $C_{\mathrm{CO-Ar}}(\nu)$ by 30% and shifts its maximum by about 10 cm^{-1} to a higher frequency region.

Table 2 The impact of the resonance spectrum model parameters on $C_{\mathrm{CO-Ar}}$

Parameter	Parameter change	Continuum change
ν_{cut}	25 → 50 cm^{-1}	−100%
	25 → 10 cm^{-1}	+15%
	25 → 1 cm^{-1}	+30%
S_{CO}	1–2%	±10–20%
$\gamma_{\mathrm{CO-CO}}$	2–5%	±5–30%
$\gamma_{\mathrm{CO-Ar}}$	0.7–1.2%	
Y	[18] → [28]	+25%
	20%	±10%

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